

Subject Code	AF5701
Subject Title	Artificial Intelligence for Business
Credit Value	3
Level	5
Pre-requisite/Exclusion	None
Objectives	This subject aims to tackle complex real-world problems with rigorous mathematical tools. In this course, students will learn the foundational principles of various artificial intelligence (AI) systems and are able to implement some basic AI techniques in their projects or other related work.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - master the basic searching techniques for problem solving and use them in game playing; - know how to represent knowledge and use them in inferences and reasoning; - manage uncertainty and reason in uncertainty situations; - design and develop knowledge based expert systems; - use basic artificial intelligence techniques to solve different data analytic problems; - understand deep learning and artificial neural networks; - participate in team work, presentation and technical writing.
Subject Synopsis/ Indicative Syllabus	• Introduction Overview; AI History (logical, neural, statistical); ethics and responsibility • Searching Techniques Modeling: defining search problems; tree search: basic exhaustive search, BFS, DFS; uniform cost search (UCS); correctness of UCS; A* search • Machine learning Overview; linear regression; linear classification; group distributional robust optimization (DRO); non-linear features; feature templates; neural networks; backpropagation; generalization; k-means • Games Overview; games modeling; game evaluation; expectimax; minimax; expectiminimax; evaluation functions; alpha-beta pruning • Large Language Models Basis, applications (text generation, image generation, audio and video generation, code generation).

Teaching/Learning Methodology	This course explores the core AI concepts. It provides a comprehensive introduction to the problems and techniques of artificial intelligence. Theory and practice are both emphasized. To enhance the understanding of how conceptions and ideas in AI are actually implemented, prolog and expert system shells will be used for programming exercises and projects. A fair portion of guided reading will also be provided. 39 hours of class activities including - lecture, tutorial, lab, workshop seminar where applicable.																																											
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% weighting</th><th colspan="7">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th></tr><tr><td>Assignments, Tests & Projects</td><td>55</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Final Examination</td><td>45</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> Assignments, Tests & Projects (55%) Practical Application and Skills Development: Assignments and projects require students to apply AI concepts and techniques to real-world or simulated business problems. This directly assesses students’ ability to master searching techniques (ILO 1), represent knowledge and perform reasoning (ILO 2), manage uncertainty (ILO 3), design expert systems (ILO 4), and use AI for data analytics (ILO 5). Teamwork, Communication, and Technical Writing: Many assignments and projects are designed to be collaborative, fostering teamwork and communication skills (ILO 7). Technical writing is assessed through project reports and documentation. Continuous Assessment: Regular tests and assignments provide ongoing feedback, allowing students to track their progress and deepen their understanding of foundational and advanced AI topics, including deep learning and neural networks (ILO 6). Authentic Assessment: By engaging students in hands-on programming exercises (e.g., using Prolog and expert system shells), these methods ensure that students not only understand theoretical concepts but can also implement and experiment with AI systems. Final Examination (45%) Comprehensive Understanding: The final exam assesses students’ overall grasp of the course material, including theoretical foundations, problem-solving strategies, and the ability to synthesize and apply knowledge across different AI domains. Individual Accountability: The exam ensures that each student individually demonstrates mastery of the learning outcomes, particularly in areas such as search algorithms, knowledge representation, reasoning under uncertainty, and understanding of deep learning (ILOs 1–6). Critical Thinking and Integration: Exam questions are designed to test higher-order thinking, requiring students to analyze, evaluate, and integrate concepts learned throughout the course.	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed							a	b	c	d	e	f	g	Assignments, Tests & Projects	55	✓	✓	✓	✓	✓	✓	✓	Final Examination	45	✓	✓	✓	✓	✓	✓		Total	100							
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Total	100																																											

Student study effort expected	Class Contact:	
	Class activities (lecture, tutorial, lab)	39 hours
	Other student study effort:	
	Assignments, Quizzes, Projects, Exams	96 hours
	Total student study effort	135 hours
Reading list and references	(1). Bratko, I., 2011, PROLOG, Programming for Artificial Intelligence, 4th edition, Addison-Wesley. (2). Luger, G.F., 2009, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th edition, Addison-Wesley. (3). Russell, S. and Norvig, P., 2021, Artificial Intelligence - A Modern Approach, 4th edition, Prentice Hall.	